📄 Website Traffic Analysis Report

1. Introduction

This project focuses on analyzing website traffic data to understand user behavior and engagement patterns. The objective is to evaluate traffic trends, identify high-performing content, analyze user interaction through events, and determine bounce behavior. The insights derived from this analysis can help improve user experience and optimize conversion strategies for Alfido Tech.

2. Data Cleaning

The dataset was cleaned to ensure consistency and accuracy. The date column was converted into datetime format to enable time-based analysis. Duplicate records were removed, and missing values were checked and handled appropriately.

It is important to note that the dataset does not contain user-level or session-level identifiers. Therefore, metrics such as total users, sessions, and average session duration could not be calculated. Additionally, referral source analysis could not be performed due to the absence of source-related data.

3. Key Performance Indicators (KPIs)

The following KPIs were calculated to evaluate website performance:

- Total Events: 122,567
- Click-Through Rate (CTR): 44.30%
- Bounce Rate: (Add your calculated value here)

These metrics provide an overview of overall traffic, user engagement, and interaction levels.

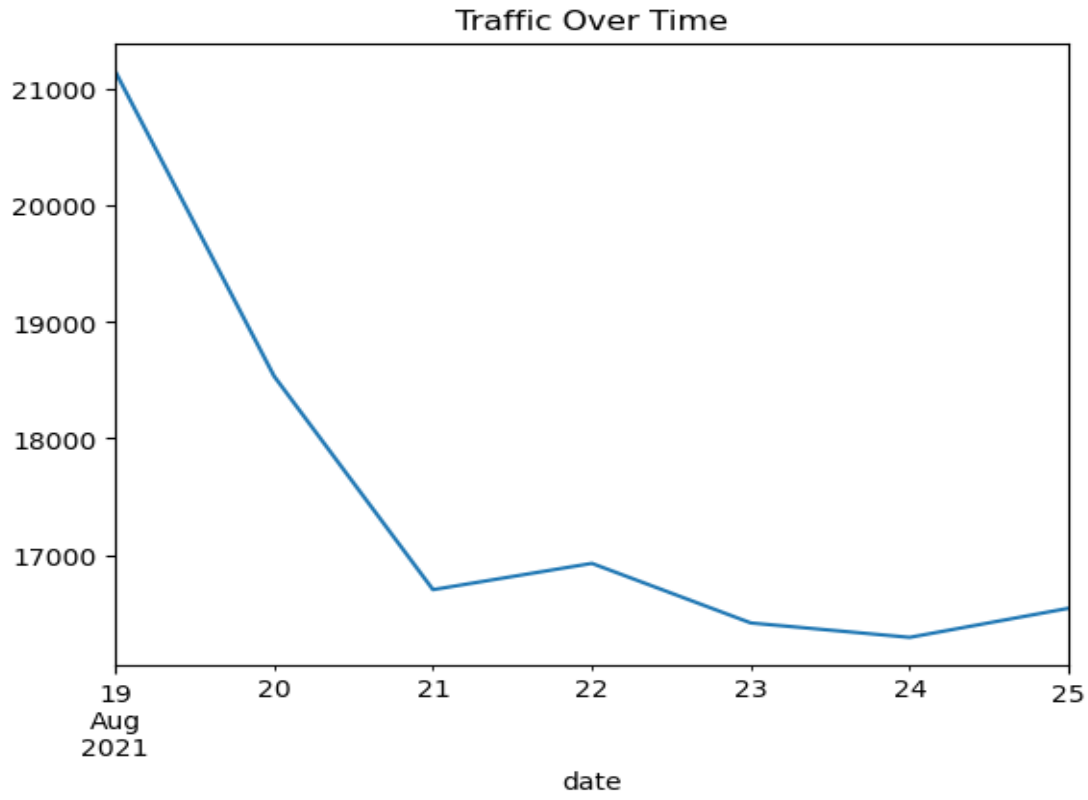
4. Data Visualization

Several visualizations were created to analyze traffic patterns and user behavior:

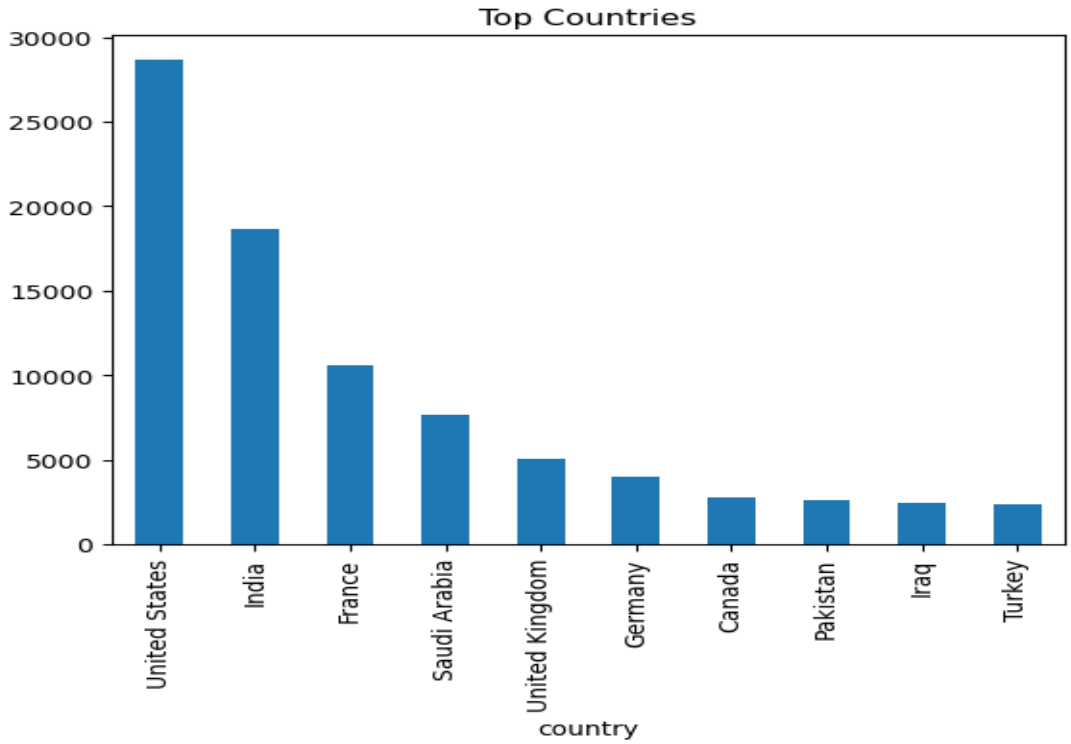
• Traffic Over Time: Showed variation in website activity across different dates
• Event Distribution: Compared pageviews, clicks, and previews
• Top Countries: Identified regions generating the most traffic
• Top Artists: Highlighted high-performing content
• Top Tracks: Showed most engaging tracks
• Top Cities: Provided location-based traffic insights

These visualizations helped in identifying patterns in user engagement and content performance.

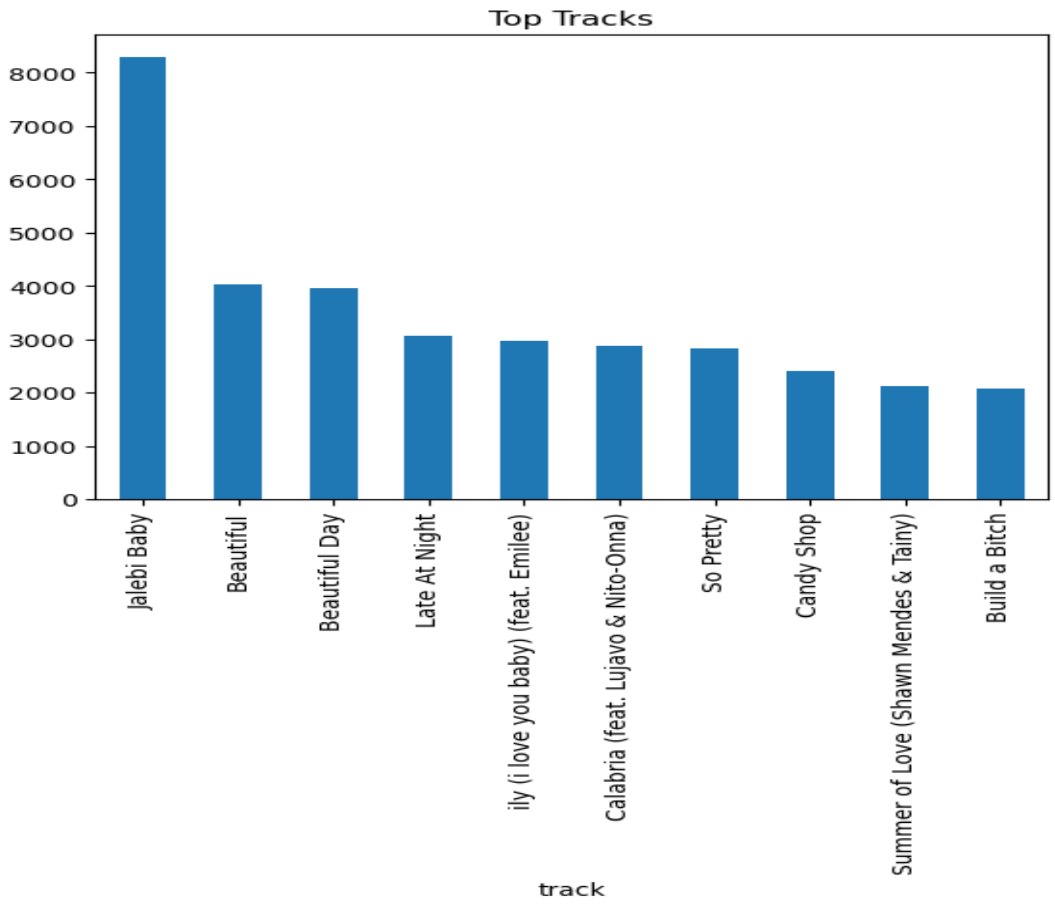
4.1 Traffic Over Time



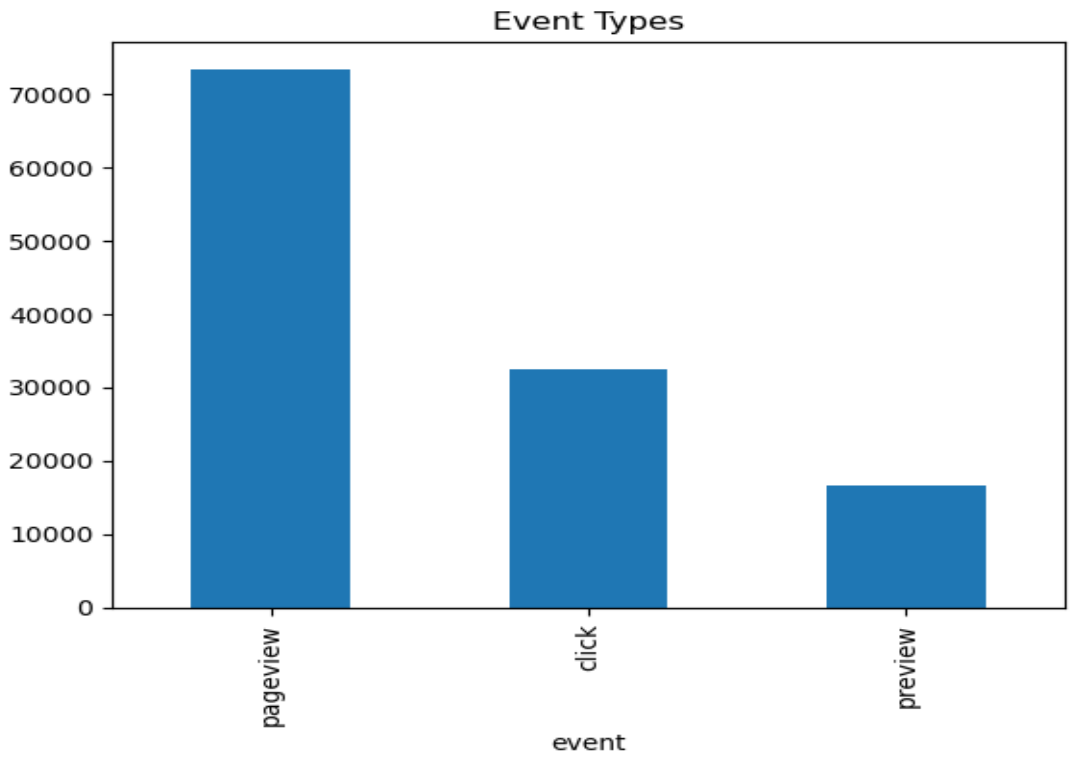
4.2 Top Countries



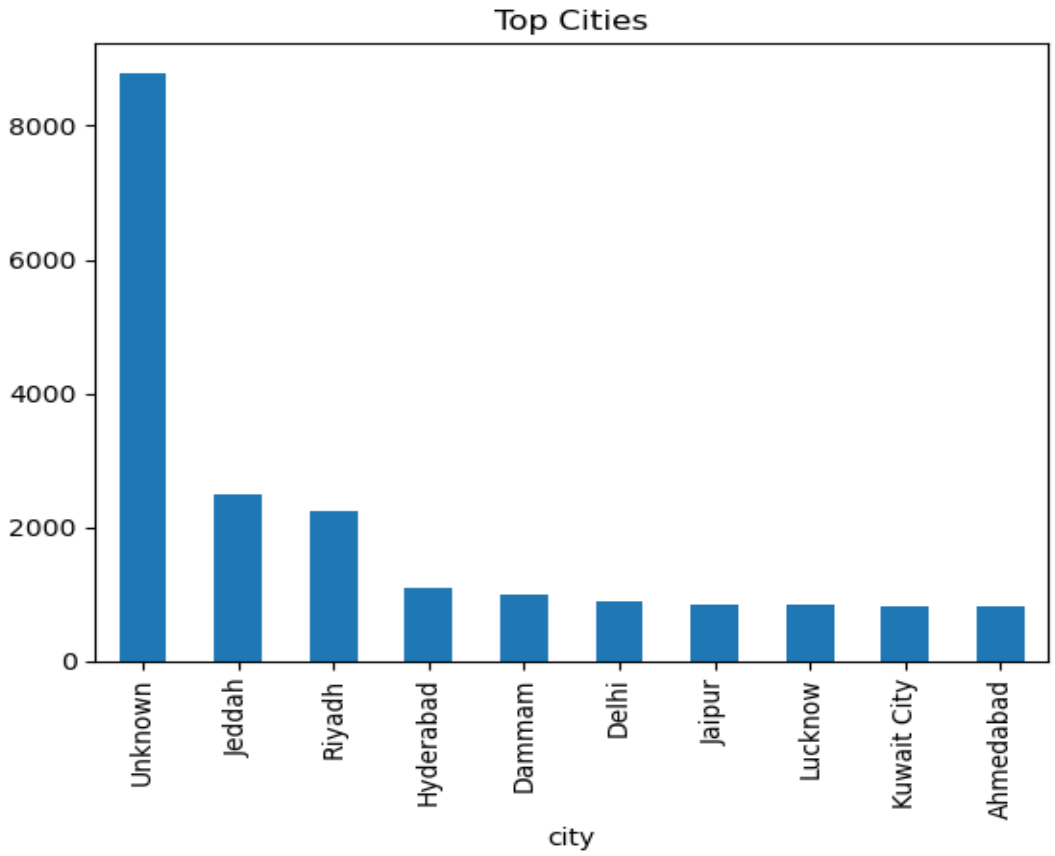
4.3 Top Tracks



4.3 Event Types



4.4 Top Cities



5. Key Insights

• Pageviews are significantly higher than clicks, indicating high traffic but moderate engagement
• The click-through rate shows that users are interacting with content, but there is scope for improvement
• A few countries contribute the majority of traffic, showing uneven distribution of users
• Certain tracks and artists receive the highest engagement, indicating popular content
• The bounce rate is moderately high, suggesting that many users leave after minimal interaction
• User journey is approximated using event types such as pageview, click, and preview

6. Recommendations

Based on the analysis, the following actions are recommended:

1. Improve call-to-action (CTA) elements to increase user clicks and engagement
2. Focus marketing efforts on high-traffic countries to maximize reach
3. Promote high-performing tracks and artists to increase user interaction
4. Optimize low-performing content to improve engagement
5. Enhance website design and user experience to reduce bounce rate

7. Conclusion

This analysis provides valuable insights into website traffic behavior and user engagement patterns. The findings highlight key areas for improvement, such as increasing interaction and reducing bounce rates. By implementing the recommended strategies, Alfido Tech can improve overall website performance and conversions.

8. Future Scope

- Include user-level and session-level data for deeper analysis
- Perform detailed user journey tracking
- Apply predictive models to forecast traffic and engagement